

Hans-Peter Marshall

Cryosphere Geophysics And Remote Sensing (CryoGARS) group
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Education

- 2005 Ph.D, Civil Engineering (Geotechnical), GPA: 3.9/4.0
Institute of Arctic and Alpine Research, University of Colorado at Boulder
Thesis: *Snowpack spatial variability: towards understanding
its effect on remote sensing measurements and snow slope stability*
- 1999 B.S., Physics with honors, Geophysics Minor (1st in UW history), GPA: 3.6/4.0
University of Washington
Thesis: *Wet snow densification*
- 1994 Nathan Hale High School, Seattle, Washington, GPA: 3.9/4.0

Professional Experience

- 08/2013-present Associate Professor, Department of Geosciences and Center for Geophysical
Investigation of the Shallow Subsurface (CGISS), Boise State University
- 07/2007-present Geophysical Engineer (Expert Consultant), U.S. Army Corps of Engineers
Cold Regions Research and Engineering Lab (CRREL), Hanover, New Hampshire
- 08/2008-07/2013 Assistant Professor, Professor, Department of Geosciences and Center for Geophysical
Investigation of the Shallow Subsurface (CGISS), Boise State University
- 04/2006-08/2008 Research Associate, Inst. of Arctic and Alpine Research (INSTAAR)
University of Colorado at Boulder
- 12/2004-04/2006 Research Geophysical Engineer (GS-11), Student Career Experiences Program (SCEP)
U.S. Army Corps of Engineers CRREL, Hanover, New Hampshire & Fairbanks, Alaska
- 01/2004-07/2004 Invited visiting scientist, Swiss Federal Institute for Snow and
Avalanche Research (SLF), Davos, Switzerland
- 07/2000-12/2005 Research Assistant, INSTAAR, University of Colorado at Boulder
- 11/2002-12/2003 Avalanche Instructor, Alpine World Ascents, Boulder, Colorado
- 06/1995-07/2000 Undergraduate Research Assistant, Glaciology Group, University of Washington
- 08/1999-03/2000 Volunteer high school teacher, Old Crow, Yukon Terr., Canada
- Summers 1993-94 Roofer, Seattle, WA

Research Interests

Snow science and glaciology, using geophysics, engineering, and remote sensing
Spatial variability of snow, snow and ice mechanics, snow slope stability, meltwater dynamics
Computational statistics, geostatistics, compressed sensing, instrumentation development

Honors and Awards

2018	NASA Robert H. Goddard Team Award: SnowEx Year 1 Organizing Team
2017	Incoming Chair of the International Snow Working Group for Remote sensing (ISWGR)
2012	Timeless Award, University of Washington, 1 of 150 College of Arts and Sciences alumni awarded from UW history on 150th yr anniv. (1861-2011), 1 of 3 from Physics Department
2010-13	New Investigator Program Award, NASA, 1st in Idaho history
2010	Young Scientist Award, Cryosphere Section, American Geophysical Union
2009	Top Reviewer Award, Cold Regions Science and Technology (CRST)
2008	U.S. Young Scientist Award, International Union of Radio Science (URSI)
2002-05	N.A.S.A. Earth Systems Science Graduate Fellowship
2003	Backcountry Access Avalanche Education Support Grant
2003	Arctic Research Consortium of the U.S. (ARCUS) Award for Arctic Research Excellence, Hon. Mention
2002	University of Colorado Graduate School Student Travel Grant
2001-02	N.S.F. K-12 Engineering Outreach Fellowship
2002	Beverly Sears Graduate Research Grant (University of Colorado)
2001	American Avalanche Association Graduate Research Grant
2000	Li Tze Fong Memorial Fellowship, University of British Columbia (declined)
1999	University of Washington Bonderman Honors Travel Fellowship
1999	Volunteer of the Year, Arctic Canada, Frontiers Foundation
1997-99	University of Washington Mary Gates Undergraduate Research Grant
1999	Featured speaker, University of Washington Undergraduate Research Symposium
1995-98	N.A.S.A. Space Grant Undergraduate Research Program
1999,2001	N.S.F. Graduate Fellowship Program - Honorable Mention
1994-95,97-99	Dean's List, University of Washington
1994-95	U.W. Undergraduate Scholar Award (1994-95)
1994	Washington Athletic Club Student Athlete of the Year, Nathan Hale High School

Student Awards

2012	Andrew Hedrick, M.S. student <i>American Avalanche Association Graduate Student Grant</i>
2012	Scott Havens, Ph.D. student <i>1st place poster NSF/NASA EPSCoR annual meeting</i>
2012	Kris Burch, Lloyd Lowe II, Kolton Drake, Undergrads, <i>Best Senior Project, BSU College of Engineering</i>
2003	Adam Maerz, High School student <i>Colorado State Science Fair Finalist</i>

Refereed Publications [h-index=20, i10-index=32, citations=1102]

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| in-review | <p>McGrath, D., Sass, L., O’Neel, S., McNeil, C., Candela, S. G., Baker, E. H., and H.P. Marshall (2018). Interannual snow accumulation variability on glaciers derived from repeat, spatially extensive ground-penetrating radar surveys. <i>The Cryosphere Discussions</i>, (doi.org/10.5194/tc-2018-126)</p> <p>Robertson, M., H.P. Marshall, Roberts, D., and Kunkel, M. (in review). Continuous snow water equivalent measurements with ultra-wideband impulse radar: A new sensor and time-delay method of processing for downward-looking radar. <i>Hydrological Processes</i></p> <p>Deeb, E., Forster, R., and H.P. Marshall (in review). Estimating snow water equivalent (SWE) using L-Band interferometric synthetic aperture radar (InSAR). <i>Remote Sensing of Environment</i></p> |
| 2018 | <p>Graeter, K. A., Osterberg, E. C., Ferris, D. G., Hawley, R. L., H.P. Marshall, Lewis, G., Meehan, T., McCarthy, F., and Birkel, S. D. (2018). Ice core records of West Greenland surface melt and climate forcing. <i>Geophysical Research Letters</i>, 45(7):3164–3172</p> <p>Kim, Y., Reck, T. J., Alonso-delPino, M., Painter, T. H., H.P. Marshall, Bair, E. H., Dozier, J., Chattopadhyay, G., Liou, K.-N., Chang, M.-C. F., and Tang, A. (2018). A Ku-band CMOS FMCW radar transceiver for snowpack remote sensing. <i>IEEE Transactions On Geoscience and Remote Sensing</i>, PP(99):1–15</p> <p>Hagenmuller, P., van Herwijnen, A., Pielmeier, C., and H.P. Marshall (2018). Evaluation of the snow penetrometer Avatech SP2. <i>Cold Regions Science and Technology</i>, 149:83–94</p> |
| 2017 | <p>Mikesell, T., van Wijk, K., Otheim, L., H.P. Marshall, and Kurbatov, A. (2017). Laser ultrasound observations of mechanical property variations in ice cores. <i>Geosciences</i>, 7(3):47</p> <p>Lewis, G., Osterberg, E., Hawley, R., Whitmore, B., H.P. Marshall, and Box, J. (2017). Regional Greenland accumulation variability from Operation IceBridge airborne accumulation radar. <i>The Cryosphere</i>, 11(2):773</p> <p>McNamara, J. P., Benner, S. G., Poulos, M. J., Pierce, J. L., Chandler, D. G., Kormos, P. R., H.P. Marshall, Flores, A. N., Seyfried, M., Glenn, N. F., and Aishlin, P. (2017). Form and function relationships revealed by long-term research in a semiarid mountain catchment. <i>WIREs Water</i>, e1267(10.1002/wat2.1267)</p> <p>Pegau, W. S., Garron, J., Zabilansky, L., Bassett, C., Bello, J., Bradford, J., Carns, R., Courville, Z., Eicken, H., Elder, B., Eriksen, P., Lavery, A., Light, B., Maksym, T., H.P. Marshall, Oggier, M., Perovich, D., Pacwiardowski, P., Singh, H., Tang, D., Wiggins, C., and Wilkinson, J. (2017). Detection of oil in and under ice. <i>International Oil Spill Conference Proceedings</i>, 2017(1):1857–1876</p> |

- 2016 Evans, S., Flores, A., Heilig, A., Kohn, M., **Marshall, H.P.**, and McNamara, J. (2016). Isotopic evidence for lateral flow and diffusive transport, but not sublimation, in a sloped seasonal snowpack, Idaho, U.S.A. *Geophysical Research Letters*, 43(7):3298–3306
- Rutter, N., **Marshall, H.P.**, Tape, K., Essery, R., and King, J. (2016). Impact of spatial averaging on radar reflectivity at internal snowpack layer boundaries. *Journal of Glaciology*, 62(236):1065–1074
- Bradford, J. H., Babcock, E. L., **Marshall, H.P.**, and Dickins, D. F. (2016). Targeted reflection-waveform inversion of experimental ground-penetrating radar data for quantification of oil spills under sea ice. *Geophysics*, 81(1):WA59–WA70
- 2015 Heilig, A., Mitterer, C., Schmid, L., Wever, N., Schweizer, J., **Marshall, H.P.**, and Eisen, O. (2015). Seasonal and diurnal cycles of liquid water in snow-measurements and modeling. *Journal of Geophysical Research-Earth Surface*, 120(10):2139–2154
- Kormos, P. R., McNamara, J. P., Seyfried, M. S., **Marshall, H.P.**, Marks, D., and Flores, A. N. (2015). Bedrock infiltration estimates from a catchment water storage-based modeling approach in the rain snow transition zone. *Journal of Hydrology*, 525:231–248
- Hedrick, A., **Marshall, H.P.**, Winstral, A., Elder, K., Yueh, S., and Cline, D. (2015). Independent evaluation of the SNODAS snow depth product using regional-scale LiDAR-derived measurements. *Cryosphere*, 9(1):13–23
- 2014 Kormos, P. R., Marks, D., McNamara, J. P., **Marshall, H.P.**, Winstral, A., and Flores, A. N. (2014a). Snow distribution, melt and surface water inputs to the soil in the mountain rain-snow transition zone. *Journal of Hydrology*, 519:190–204
- Walters, R. D., Watson, K. A., **Marshall, H.P.**, McNamara, J. P., and Flores, A. N. (2014). A physiographic approach to downscaling fractional snow cover data in mountainous regions. *Remote Sensing of Environment*, 152:413–425
- Havens, S., **Marshall, H.P.**, Johnson, J. B., and Nicholson, B. (2014b). Calculating the velocity of a fast-moving snow avalanche using an infrasound array. *Geophysical Research Letters*, 41(17):6191–6198
- Anderson, B. T., McNamara, J. P., **Marshall, H.P.**, and Flores, A. N. (2014). Insights into the physical processes controlling correlations between snow distribution and terrain properties. *Water Resources Research*, 50(6):4545–4563
- Tinkham, W. T., Smith, A. M. S., **Marshall, H.P.**, Link, T. E., Falkowski, M. J., and Winstral, A. H. (2014). Quantifying spatial distribution of snow depth errors from lidar using random forest. *Remote Sensing of Environment*, 141:105–115

- McCreight, J. L., Slater, A. G., **Marshall, H.P.**, and Rajagopalan, B. (2014). Inference and uncertainty of snow depth spatial distribution at the kilometre scale in the colorado rocky mountains: the effects of sample size, random sampling, predictor quality, and validation procedures. *Hydrological Processes*, 28(3):933–957
- Kormos, P. R., Marks, D., Williams, C. J., **Marshall, H.P.**, Aishlin, P., Chandler, D. G., and McNamara, J. P. (2014b). Soil, snow, weather, and sub-surface storage data from a mountain catchment in the rain-snow transition zone. *Earth System Science Data*, 6(1):165–173
- 2013 Havens, S., **Marshall, H.P.**, Pielmeier, C., and Elder, K. (2013b). Automatic grain type classification of snow micro penetrometer signals with random forests. *IEEE Transactions On Geoscience and Remote Sensing*, 51(6):3328–3335
- Eiriksson, D., Whitson, M., Luce, C. H., **Marshall, H.P.**, Bradford, J., Benner, S. G., Black, T., Hetrick, H., and McNamara, J. P. (2013). An evaluation of the hydrologic relevance of lateral flow in snow at hillslope and catchment scales. *Hydrological Processes*, 27(5):640–654
- Tinkham, W. T., Hoffman, C. M., Falkowski, M. J., Smith, A., **Marshall, H.P.**, and Link, T. E. (2013a). A methodology to characterize vertical accuracies in lidar-derived products at landscape scales. *Photogrammetric Engineering & Remote Sensing*, 79(8):709–716
- 2012 Mikesell, T. D., van Wijk, K., Haney, M. M., Bradford, J. H., **Marshall, H.P.**, and Harper, J. T. (2012). Monitoring glacier surface seismicity in time and space using rayleigh waves. *Journal of Geophysical Research-Earth Surface*, 117(F2):F02020
- 2011 **Marshall, H.P.**, Schweizer, J., and Birkeland, K. (2011). Some recent advances in snow and avalanche science. *Cold Regions Science and Technology*, 69(2-3):119–121
- Geroy, I., Gribb, M., **Marshall, H.P.**, Chandler, D., Benner, S. G., and McNamara, J. P. (2011). Aspect influences on soil water retention and storage. *Hydrological Processes*, 25(25):3836–3842
- 2010 Tape, K. D., Rutter, N., **Marshall, H.P.**, Essery, R., and Sturm, M. (2010b). Recording microscale variations in snowpack layering using near-infrared photography. *Journal of Glaciology*, 56(195):75–80
- Tape, K. D., Lord, R., **Marshall, H.P.**, and Ruess, R. W. (2010a). Snow-mediated ptarmigan browsing and shrub expansion in arctic alaska. *Ecoscience*, 17(2):186–193
- Koh, G., Lever, J., Arcone, S., **Marshall, H.P.**, and Ray, L. (2010). Autonomous fmcw radar survey of antarctic shear zone. *IEEE Publications, 13th International Conference on Ground Penetrating Radar*, DOI: 10.1109/ICGPR.2010.5550174
- 2009 **Marshall, H.P.** and Johnson, J. B. (2009). Accurate inversion of high-resolution snow penetrometer signals for microstructural and micromechanical properties. *Journal of Geophysical Research-Earth Surface*, 114(F04016)

- Lutz, E., Birkeland, K., and **Marshall, H.P.** (2009b). Quantifying changes in weak layer microstructure associated with artificial load changes. *Cold Regions Science and Technology*, 59(2-3):202–209
- Pielmeier, C. and **Marshall, H.P.** (2009). Rutschblock-scale snowpack stability derived from multiple quality-controlled snowmicropen measurements. *Cold Regions Science and Technology*, 59(2-3):178–184
- 2008 **Marshall, H.P.** and Koh, G. (2008). FMCW radars for snow research. *Cold Regions Science and Technology*, 52(2):118–131
- Hardy, J., Davis, R., Koh, Y., Cline, D., Elder, K., Armstrong, R., **Marshall, H.P.**, Painter, T., Saint-Martin, G. C., Deroo, R., Sarabandi, K., Graf, T., Koike, T., and McDonald, K. (2008). NASA Cold Land Processes Experiment (CLPX 2002/03): Local Scale Observation Site. *Journal of Hydrometeorology*, 9(6):1434–1442
- 2007 **Marshall, H.P.**, Schneebeli, M., and Koh, G. (2007). Snow stratigraphy measurements with high-frequency FMCW radar: Comparison with snow micro-penetrometer. *Cold Regions Science and Technology*, 47(1-2):108–117
- O’Neel, S., **Marshall, H.P.**, McNamara, D. E., and Pfeffer, W. T. (2007). Seismic detection and analysis of icequakes at Columbia Glacier, Alaska. *Journal of Geophysical Research-Earth Surface*, 112(F3)
- Waddington, E. D., Neumann, T. A., Koutnik, M. R., **Marshall, H.P.**, and Morse, D. L. (2007). Inference of accumulation-rate patterns from deep layers in glaciers and ice sheets. *Journal of Glaciology*, 53(183):694–712
- 2005 **Marshall, H.P.**, Koh, G., and Forster, R. R. (2005). Estimating alpine snowpack properties using FMCW radar. *Annals of Glaciology*, 40:157–162
- 2004 **Marshall, H.P.**, Koh, G., and Forster, R. R. (2004). Ground-based frequency-modulated continuous wave radar measurements in wet and dry snowpacks, Colorado, USA: an analysis and summary of the 2002-03 NASA CLPX data. *Hydrological Processes*, 18(18):3609–3622
- 2002 **Marshall, H.P.**, Harper, J. T., Pfeffer, W. T., and Humphrey, N. F. (2002). Depth-varying constitutive properties observed in an isothermal glacier. *Geophysical Research Letters*, 29(23):61–1:61–4
- 1999 **Marshall, H.P.**, Conway, H., and Rasmussen, L. A. (1999). Snow densification during rain. *Cold Regions Science and Technology*, 30(1-3):35–41
- Conway, H., Rasmussen, L. A., and **Marshall, H.P.** (1999). Annual mass balance of Blue Glacier, USA: 1955-97. *Geografiska Annaler, Ser. A-Phys. Geogr.*, 81A(4):509–520

Morse, D. L., Waddington, E. D., **Marshall, H.P.**, Neumann, T. A., Steig, E. J., Dibb, J. E., Winebrenner, D. P., and Arthern, R. J. (1999). Accumulation rate measurements at Taylor Dome, East Antarctica: Techniques and strategies for mass balance measurements in polar environments. *Geografiska Annaler, Ser. A-Phys. Geogr.*, 81A(4):683–694

Refereed Book Chapters / Technical Reports

- 2015 | **Marshall, H.P.**, Hawley, R., and Tedesco, M. (2015). *Field measurements for remote sensing of the cryosphere*, chapter 14, pages 345–381. Remote Sensing of the Cryosphere. John Wiley & Sons, Ltd, ed. M. Tedesco
- Arenson, L., Colgan, W., and **Marshall, H.P.** (2015). *Physical, Thermal, and Mechanical Properties of Snow, Ice, and Permafrost*, chapter 2, pages 35–75. Snow and ice-related hazards, risks, and disasters. Elsevier, ed. Haeberli and Whiteman, Radarweg 29, PO Box 211, 1000 AE Amsterdam, Netherlands
- 2014 | Havens, S., **Marshall, H.P.**, Trisca, G., and Johnson, J. B. (2014c). Real time avalanche detection for high risk areas. Monograph FHWA-ID-14-219, Idaho Transportation Department, 3311 W State Street, P. O. Box 7129, Boise, ID 83707-1129 United States
- 2013 | Rott, H., Nagler, T., Prinz, R., Voglmeier, K., Fromm, R., Adams, M. S., Yueh, S., Elder, K., **Marshall, H.P.**, Coccia, A., Imbembo, E., Schuttemeyer, D., and Kern, M. (2013). AlpSAR 2012-13, a field experiment on snow observations and parameter retrievals with Ku-and X-band radar. Technical Report 722, European Space Agency Special Publication

Magazine Articles / Podcasts

- 2013 | Durham, T., **Marshall, H.P.**, Tsang, L., Racette, P., Bonds, Q., Miranda, F., and Vanhille, K. (2013). Wideband sensor technologies for measuring surface snow. *Earthzine*, December 2(earthzine.org)
- 2012 | **Marshall, H.P.** (2012). Researching snow: More than just hitting the slopes. Podcast <https://beyondtheblue.boisestate.edu/blog/2012/01/09/hp-marshall/>, Beyond the Blue: Boise State Faculty Podcasts
- 2011 | Sturm, M. and **Marshall, H.P.** (2011). Marshall receives Cryosphere Young Investigator Award (citation and response). *EOS Transactions, AGU*, 92(22):189

Archived Datasets

- 2018 | Elder, K., Brucker, L., Hiemstra, C., and **H.P. Marshall** (2018). Snowex17 community snow pit measurements, version 1. Digital media, <https://doi.org/10.5067/Q0310G1XULZS>, NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA

- Brucker, L., Hiemstra, C., **H.P. Marshall**, and Elder, K. (2018). Snowex17 community snow depth probe measurements, version 1. Digital media, <https://doi.org/10.5067/WKC6VFMT7JTF>, NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA
- 2004 **Marshall, H.P.**, Koh, G., and Forster, R. R. (2004). Ground-based frequency-modulated continuous wave radar measurements in wet and dry snowpacks, Colorado, USA: an analysis and summary of the 2002-03 NASA CLPX data. *Hydrological Processes*, 18(18):3609–3622

Research Funding Summary

Agency	Short Title	Role	Dates	Amount
NASA	NASA SnowEx Ground-based Radar	Co-PI	6/2018-6/2021	\$444,742
NASA	NASA SnowEx Y1 Data Assimilation	Co-PI	6/2018-6/2021	\$421,212
NASA	Spatiotemporal Patterns in GBRs	PI	6/2017-6/2022	\$460,719
NASA	L-Band InSAR for snow depth and SWE	Co-PI	6/2017-6/2022	\$450,000
NASA JPL	ASO and UAVSAR data fusion	PI	6/2017-7/2018	\$21,103
U.S. DOTs	Infrasound Avalanche Detection	PI	6/2017-8/2018	\$99,339
NASA	Broadband Array IIP - Phase 2	PI	02/2015-05/2017	\$330,000
NSF	GreenTRACS	PI	01/2015-12/2017	\$355,338
NASA	Multi-frequency Data Assimilation	Co-PI	07/2015-06/2018	\$295,577
NASA	Monitoring Earth's Hydrosphere	Co-PI	06/2014-05/2017	\$749,870
U.S. Army	SWE estimates using radar phase	Co-PI	10/2013- 9/2016	\$490,800
OGP	Remote Sensing Tests with Oil in Ice	Co-PI	01/2014-12/2014	\$128,739
NASA	NASA Snow School	Co-PI	01/2014-12/2014	\$65,385
NSF	MRI: Laser-ultrasonic Ice Core Tomography	Co-PI	09/2012-08/2013	\$336,122
NSF	MRI:GPU Computing/Visualization Cluster	Co-PI	09/2012-08/2013	\$555,384
NSF	Measuring Sublimation with Isotopes	Co-PI	04/2012-04/2013	\$19,912
IDT	Avalanche Detection	PI	09/2012-08/2015	\$100,000
NASA	WISM IIP - Phase 1	PI	11/2011-12/2014	\$94,947
NASA	Remote Sensing of the Cryosphere	PI	01/2011-12/2013	\$1,124,552
NASA	Improvement of Remotely Sensed SWE	PI	09/2010-08/2013	\$329,953
NSF	Lateral Flow of Water in Snow	PI	04/2010-03/2013	\$228,963
NASA	Earth Systems Science Fellowship	PI	09/2010-12/2013	\$90,000
D.F. Dickins	Oil Detection Radar	PI	01/2010-12/2012	\$328,747
IDT	Avalanche Modeling	PI	09/2010-08/2012	\$10,000
NASA	CLPX-II Grand Mesa, Colorado	PI	08/2010-08/2012	\$35,468
NASA	Investigation of Snowpack Properties	PI	08/2006-07/2009	\$462,132
NSF	Radar for Measuring Water Pathways	PI	09/2005-08/2010	\$69,785
			TOTAL	\$7,999,450

Awarded Research Grants

6/2018-6/2021 NASA Terrestrial Hydrology Program 2017

Constraining airborne and satellite algorithms for estimating snow water equivalent through spatially extensive ground penetrating radar observations during NASA's SnowEx campaign
Co-PI, \$444,742

PI: Dan McGrath (Colorado State Univ.), Co-PI: Ryan Webb (Univ. Colorado)

6/2018-6/2021 NASA Terrestrial Hydrology Program 2017

Using SnowEx Year 1 Data to Assess and Improve Snow Measurements, Validation, Algorithms, and Data Assimilation in Forest and Shrubland Landscapes

Co-PI, \$421,212

PI: Chris Hiemstra (CRREL), Co-PIs: Kelly Elder (USFS), Ludo Brucker (NASA Goddard)

6/2017-6/2022 NASA Terrestrial Hydrology Program 2016

Spatiotemporal patterns in ground-based snow remote sensing and in-situ observations: Analysis of CLPX-II, CoreH2O, and SnowEx Y1 data for sampling design and site selection

Principal Investigator, \$460,719

Co-PIs: Kelly Elder (USFS), Eli Deeb (CRREL), Ludo Brucker (NASA Goddard)

6/2017-6/2022 NASA Terrestrial Hydrology Program 2016

Remote sensing strategies and requirements for L-Band interferometric snow depth and snow water equivalent estimation

Co-PI, \$450,000

PI: Eli Deeb (CRREL), Co-PIs: Rick Forster (Univ. Utah), Cathleen Jones (NASA JPL), Chris Hiemstra (CRREL)

6/2017-8/2018 NASA Jet Propulsion Laboratory, CalTech

ASO and UAVSAR Data Fusion: Exploring the Advantages of Combining Airborne LIDAR and InSAR

Principal Investigator, \$21,103

6/2017-8/2018 Consortium of U.S. Department of Transportation Avalanche Forecasting Programs

Infrasound Avalanche Detection: Comparison Study of Multiple New Technologies

Principal Investigator, \$99,339

Co-PIs: Jeff Johnson, Sin Ming Loo (BSU)

2/2015-5/2017 NASA Instrument Incubator Program (IIP) - Phase 2

Broadband Array Technology in Support of the Snow and Cold Land Process (SCLP) Mission

Principal Investigator, \$330,000

1/2015-12/2017 NSF Office of Polar Programs

Collaborative Research: GreenTrACS: Greenland Traverse for Accumulation and Climate Studies

Principal Investigator, \$355,338

Co-PI: John Bradford (BSU)

7/2015-6/2018 NASA Terrestrial Hydrology Program (THP)

Multiple Frequency Active Microwave Remote Sensing for Snow Water Equivalent Retrieval from Space: A Data Assimilation Approach

Co-PI, \$295,577

PI: Lejo Flores (BSU)

6/2014-6/2018 NASA Experimental Program to Stimulate Competitive Research (EPSCoR)

Monitoring Earth's Hydrosphere: Integrating Remote Sensing, Modeling, and Verification

Co-PI, \$749,870

PI: Lejo Flores, Co-PIs: Jim McNamara, Nancy Glen (BSU)

10/2013-9/2016 U.S. Army Basic Research Program

Quantifying the accuracy of Snow Water Equivalent estimates using radar phase over a wide range of frequencies and varying snow conditions

Co-PI, \$490,800

PI: Eli Deeb (CRREL), Co-PI: Steve Arcone (CRREL)

1/2014-12/2014 NASA Terrestrial Hydrology Program (THP)

NASA Snow School

Co-PI, \$65,340

PI: Mark Sereze (National Snow and Ice Data Center), Co-PIs: Michael Durand (Ohio State University), Kelly Elder (USFS), Jessica Lundquist (University of Washington), Noah Molotch (University of Colorado), Matthew Sturm (University of Alaska, Fairbanks)

01/2014-12/2014 International Association of Oil and Gas Producers (OGP)

Program Design and Management for Remote Sensing Tests with Oil in Ice

Co-PI, \$1,409,449 total, BSU portion: \$128,731

BSU PI: John Bradford

Collaborators: Oil Spill Recovery Institute, University of Alaska Fairbanks, Prince William Sound Science Center, U.S. Army Cold Regions Research and Engineering Laboratory (CRREL), Polar Ocean Services, Woods Hole Oceanographic Institute, Norbit, Inland Gulf Marine, University of Washington

09/2012-08/2013 NSF Major Research Instrumentation

MRI: Acquisition of a GPU-accelerated High Performance Computing and Visualization Cluster

Co-PI, \$555,384

PI: Inanc Senocak, Co-PIs: Peter Mullner, Jodi Mead, Tim Anderson (BSU)

09/2012-08/2013 NSF Major Research Instrumentation

MRI: Development of a laser-ultrasonic ice core tomography system

Co-PI, \$336,122

PI: Kasper Van Wijk (BSU)

04/2012-04/2013 NSF Hydrologic Sciences

RAPID: An unusual opportunity to track snow ablation using stable isotope evolution of the 2011-2012 snowpack near Boise, Idaho

Co-PI, \$19,912

PI: Sam Evans, Co-PIs: Alejandro Flores, Matt Kohn (BSU)

09/2012-08/2015 Idaho Department of Transportation

Development of an operational infrasonic avalanche detection system for Highway 21, Idaho

Principal Investigator, \$100,000

Co-PIs: Scott Havens, Jeff Johnson (BSU)

01/2011-12/2013 NASA Experimental Program to Stimulate Competitive Research (EPSCoR)

Remote sensing of the Cryosphere: calibration and validation

Science Principal Investigator, \$1,124,552 (\$374,932 university match)

Co-PI: John Bradford, Kasper Van Wijk, Jim McNamara, and Alejandro Flores, BSU Geosciences (BSU)

09/2010-08/2013 NASA New Investigator Program

Improvement of remotely sensed snow water equivalent estimates from microwave radar

Principal Investigator, \$329,953 (1st in Idaho history)

04/2010-03/2013 NSF Hydrological Sciences Program

Quantifying lateral flow of water in alpine snowpacks using high-resolution geophysical techniques

Principal Investigator, \$228,963

Co-PI: John Bradford and Jim McNamara, BSU

09/2010-12/2013 NASA Earth and Space Science Fellowship

Characterizing snow microstructure with X-ray tomography and a snow micropenetrometer: understanding important effects of microstructure on remote sensing

Principal Investigator, \$90,000

Student: Scott Havens

01/2010-12/2012 DF Dickins Associates and consortium of oil companies

Detecting oil on and under sea ice using ground-penetrating radar: proposal for developing a new airborne system

Principal Investigator, \$328,747

Co-PI: John Bradford, BSU

09/2010-08/2011 Idaho Department of Transportation Research Program

Development of an avalanche forecasting model for Highway 12 and 21

Principal Investigator, \$10,000

11/2011-12/2014 NASA Instrument Incubator Program (IIP) - Phase 1

An 8-40 GHz Wideband Instrument for Snow Measurements (WISM)

Principal Investigator for BSU subcontract, BSU portion \$94,947

Collaborative proposal with Harris Corporation, NASA Goddard,
University of Washington, Nuvotronics LLC

08/2010-08/2012 NASA Terrestrial Hydrology Program

Ground-based radar calibration / validation measurements for

NASA CLPX-II 2010, Grand Mesa, Colorado

Principal Investigator, \$35,468

08/2006-07/2009 NASA Energy and Water-cycle Sponsored (NEWS) Research
and Terrestrial Hydrology Program (THP)

*Investigation of Spatial and Temporal Variation in Snowpack Properties
Using Ground-Based High Resolution Microwave Radar Combined with
Detailed Snowpack Characterization*

Principal Investigator, \$462,132 (\$114,232 moved to BSU when hired)

Co-I: Nick Rutter (Northumbria University, U.K), Rick Forster (University of Utah)

09/2005-08/2010 NSF Arctic Sciences Program

*Collaborative Research: Polarimetric Characteristics of Radio-wave Scattering from Water
Pathways within Glaciers: Laboratory Experiments and Computer Simulations*

Principal Investigator, \$69,785

Co-PI: Kenichi Matsuoka (University of Washington)

Primary Advising

Name	Degree	Advisor Role	Program/University	Graduation
Achim Heilig	Post-Doc	Primary	Geophysics/BSU	01/11-12/12
Andrew Hedrick	Ph.D.	Primary	Geophysics/BSU	exp. Spring 2019
Joel Gongora	Ph.D.	Primary	Geophysics/BSU	exp. Spring 2019
Scott Havens	Ph.D.	Primary	Geophysics/BSU	Fall 2014
Hank Hetrick	Ph.D.	Primary	Geophysics/BSU	exp. Summer 2018
Chago Rodriguez	Ph.D.	Primary	Geophysics/BSU	exp. Spring 2019
Megan Mason	M.S.	Primary	Geophysics/BSU	exp. Fall 2019
Tate Meehan	M.S.	Co-primary	Geophysics/BSU	Spring 2018
Andrew Hedrick	M.S.	Primary	Geophysics/BSU	Fall 2013
Jesse Dean	M.S.	Primary	Geophysics/BSU	Fall 2016
Mark Robertson	M.S.	Primary	Geophysics/BSU	Fall 2016
Gabriel Trisca	M.S.	Primary	Computer Science/BSU	Spring 2014
Peter Olsoy	Undergrad	Primary	Biology/BSU	Spring 2011
Jennifer Morris	Undergrad	Primary	Geosciences/BSU	Spring 2011
Jason Jennings	Undergrad	Primary	Geophysics/BSU	Spring 2012
Andrew Peterson	Undergrad	Primary	Geophysics/BSU	Spring 2012
Jake Robertson	Undergrad	Primary	Geosciences/BSU	Spring 2013
Rachel Parsons	Undergrad	Primary	Geosciences/BSU	Spring 2013
Mark Robertson	Undergrad	Primary	Geophysics/BSU	Fall 2012
Andrew Karlson	Undergrad	Primary	Geosciences/BSU	Spring 2016
Robert Florence	Undergrad	Primary	Geosciences/BSU	Spring 2015
Kris Burch	Undergrad	Primary	ECE/BSU	Spring 2012
Lloyd Lowe II	Undergrad	Primary	ECE/BSU	Spring 2012
Kolton Drake	Undergrad	Primary	ECE/BSU	Spring 2012

Committee Member / Secondary Advising

Name	Degree	Advisor Role	Program/University	Graduation
Miguel Aguayo	Ph.D.	Committee	Geosciences/BSU	exp. Summer 2018
Katelyn Watson	Ph.D.	Committee	Geosciences/BSU	exp. Summer 2018
Jake Anderson	Ph.D.	Committee	Geophysics/BSU	exp. Summer 2018
Kara Ferguson	M.S.	Committee	Geosciences/BSU	exp. Spring 2018
Zakk Hess	M.S.	Committee	Geosciences/BSU	exp. Fall 2018
Zach Uhlmann	M.S.	Committee	Geosciences/BSU	exp Summer 2018
Megan Gallagher	M.S.	Committee	Geophysics/BSU	exp Summer 2018
William Rudisill	M.S.	Committee	Geosciences/BSU	exp Spring 2018
Lacie Rasley	M.S.	Committee	Geosciences/BSU	exp Spring 2018
Eric Lutz	Ph.D.	Committee	Geosci./Montana State Univ.	Spring 2009
James McCreight	Ph.D.	Committee	Civil Eng./Univ. Colorado	Fall 2011
Dylan Mikesell	Ph.D.	Committee	Geophysics/BSU	Fall 2011
Joel Brown	Ph.D.	Committee	Geophysics/BSU	Fall 2011
Eli Deeb	Ph.D.	Committee	Geography/Univ. Utah	Fall 2012
Thomas Blum	Ph.D.	Committee	Geophysics/BSU	Spring 2013
Patrick Kormos	Ph.D.	Committee	Geosciences/BSU	Fall 2013
Wade Tinkham	Ph.D.	Committee	Forest Ecol., Biogeosci./U of I	Spring 2013
Nicolas Kinar	Ph.D.	External Examiner	Hydrology, U of Saskatchewan	Fall 2013
Esther Babcock	Ph.D.	Committee	Geophysics/BSU	Spring 2015
Brian Terbush	M.S.	Committee	Geosciences/BSU	Spring 2015
Ravi Preeshageetha	M.S.	Committee	Computer Science/ BSU	Spring 2015
Josh Nichols	M.S.	Committee	Geophysics/BSU	Spring 2010
Emily Park	M.S.	Committee	Geophysics/BSU	Spring 2011
Alden Shallcross	M.S.	Committee	Geosciences/BSU	Fall 2011
Brian Anderson	M.S.	Committee	Geosciences/BSU	Spring 2011
David Eiriksson	M.S.	Committee	Geosciences/BSU	Spring 2012
Reggie Walters	M.S.	Committee	Geosciences/BSU	Fall 2013
Erik Boe	M.S.	Committee	Geosciences/BSU	Fall 2013

Teaching - BSU Courses Taught

Course Number/Name	Semester	Credits	# grad/ugrad
GEOS 486: Geosciences Capstone	Spring 2016	4	0/1
GEOPH 566: Snow and Ice Physics	Spring 2016	3	4/0
GEOPH 567: Snow Science Field Methods	Spring 2016	2	1/0
GEOPH 422/522: Data Analysis and Geostatistics	Fall 2014	3	12/1
GEOPH 502: Processes and Properties of the Earth II	Spring 2014	4	4/0
GEOPH 467/567: Snow Science Field Methods	Spring 2014	2	1/1
GEOPH 480: Research in Geophysics	Spring 2014	3	0/1
GEOPH 422/522: Data Analysis and Geostatistics	Fall 2013	3	16/0
GEOPH 502: Properties and Processes of the Earth II	Spring 2013	4	5/0
GEOPH 467/567: Snow Science Field Methods	Spring 2013	2	2/1
GEOPH 466/566: Snow and Ice Physics	Spring 2012	3	9/4
GEOPH 480: Research in Geophysics	Spring 2012	2	0/2
GEOPH 422/522: Data Analysis and Geostatistics	Fall 2011	3	9/2
GEOPH 497/597: Snow Radar Remote Sensing	Fall 2011	1	7/2
GEOPH 480: Research in Geophysics	Fall 2011	2	0/2
GEOPH 497/597: Snow Science Field Methods	Spring 2011	2	5/7
GEOPH 497/597: Data Analysis and Geostatistics	Fall 2010	3	11/2
GEOPH 497/597: Snow and Ice Physics	Spring 2010	3	7/2
GEOPH 497/597, CE 497/597: Data Analysis / Geostat.	Fall 2009	3	18/1
GEOPH 305: Applied Geophysics, w/ John Bradford	Spring 2009	3	0/5
TOTAL		55	111/34

Teaching - Other Courses

01/2003-06/2003 T.A., *The Weather*, Program on Ocean and Atmospheric Sciences, Univ. CO

01/2000-06/2000 T.A., *Engineering Geology*, Dept. of Civil Engineering, Univ. CO

12/2002-12/2003 Instructor, *Avalanche Level I*, Alpine World Ascents

08/2001-06/2002 N.S.F. K-12 Outreach Fellow

Provided lectures/hands-on demonstrations to local 7th and 8th grade classes on glaciology and avalanche related topics; communicated via website with students during field season in W. Antarctica. Supervised high-school student on avalanche-related senior research project.

09/1999-03/2000 Volunteer High School Teacher, Old Crow, Yukon Territory

Developed and taught individualized curriculum for at-risk high school youth in this small isolated Arctic village of Gwitch'in First Nation people. Supervised construction on Youth Center, obtained funding for and organized week-long hunting and cultural activities field trip, created afterschool and evening basketball program. (see also Honors / Awards)

University Service

08/2012-12/2015	Graduate Program Committee, Geophysics Rep.
2013	Chair, hiring committee, Geophysics Faculty
09/2008-09/2011	Undergraduate Curriculum Committee, Geophysics Rep.
2012	Hiring committee, COAS Scientific Instrumentation Shop, Geosciences rep.
2011,2014	Hiring committee, Geophysics Faculty position, Geoscience department
Fall 2010	Organized installation of INL software in Mini-Cave 3-D Visualization Laboratory
01/09-present	Established geodetic/survey grade GPS instrument pool for department
2010	Co-PI, BSU designated a NVIDIA CUDA HPC research center (one of 20 worldwide)
2011-present	Organize avalanche training and maintain pool of avalanche safety equipment for BSU

External Service

2016	NASA Earth Venture Mission Panel Member (proposals for future Earth observation satellites)
01/2015-present	Education board member, NASA international Snow Working Group for Remote Sensing (iSWGR)
02/2015, 01/2016	Instructor, NASA iSWGR field school
12/2008-present	Session Chair, Fall AGU Meeting (16 total sessions 2008-2012)
Jan 6-8, 2012	Co-organizer, <i>SEG Cryosphere Geophysics Workshop: Understanding a Changing Climate with Subsurface Imaging</i> , Boise, ID
12/2008-01/2012	Cryosphere Section Representative, Fall AGU meeting <i>2 trips per year to Washington, D.C. to organize Cryosphere program (> 500 submissions annually) for largest geophysics meeting in U.S. (>15,000 submissions)</i>
2011	NASA Senior Review Panel Member (review of all 13 active Earth Science NASA Missions)
2010-2011	Lead Guest Editor: CRST special issue for the <i>2010 International Snow Science Workshop</i> (premier snow science meeting worldwide)
12/2010	Invited Keynote Speaker at the PRYN/APECS Career Development Workshop <i>Involving the next generation in Cryospheric Sciences: Getting advising and mentoring experience during graduate school through participation in undergraduate and high school research programs</i>

03/2008,04/2009	Co-organized and taught Snow Characterization Workshop with Dr. Martin Schneebeli, Swiss Federal Institute for Snow and Avalanche Research (SLF) <i>3 day classroom and field-based workshop held in Silverton, Colorado in 2008 at the Center for Snow and Avalanche Studies, 10 invited graduate students and scientists from different U.S. universities. Invited to repeat workshop in 2009 in Manali, India, by the Indian Department of Defense Snow and Avalanche Study Establishment (SASE), more than 30 Indian Ph.D. scientists attended.</i>
01/2005-08/2011	Chair of Research Committee, American Avalanche Association <i>Administered four grants per year, only funding source specifically for snow avalanche research in U.S.</i>
01/2004-present	Journal Reviewer: <i>Cold Regions Science and Technology; Hydrologic Processes; Journal of Geophysical Research - Earth Surface; Journal of Geophysical Research - Atmospheres; Annals of Glaciology; Water Resources Research; Earth Surface Processes and Landforms; Journal of Glaciology; Transactions on Geoscience and Remote Sensing; Remote Sensing of Environment; Arctic, Antarctic, and Alpine Research; Geoscience and Remote Sensing Letters; Near Surface Geophysics; Northwest Science; Journal of Applied Geophysics</i>
04/2008-present	Proposal Reviewer: <i>NASA Terrestrial Hydrology Program (panel member); NSF Experimental Program to Stimulate Competitive Research (EPSCoR) program; NSF Arctic Sciences; NASA Cryospheric Sciences IceBridge (panel member); NSF Geophysics; NSF Hydrologic Sciences; American Avalanche Association</i>
2010	Organized and delivered <i>Snow Day</i> at the Discovery Center of Idaho
2011-present	Train volunteers and develop curriculum for Bogus Basin <i>Snow School</i>
06/2007-09/2009	Research Committee, Eastern Snow Conference
10/2009-09/2012	Research committee, American Institute for Avalanche Research and Education
09/2006	Session Chair, Fracture Mechanics, International Snow Science Workshop <i>also reviewed papers and helped plan this international conference; more than 800 total attendees</i>
2001-2005	Senior Research Project Mentor for 6 Boulder Valley HS students <i>Awards: 2002: Earth Sciences 1st Place, State and Nat. Finalist, Best in Show (Adam Maerz); 2001: Earth Sciences 3rd Place (John Powers)</i>

Conference Proceedings Papers

2014	Marshall, H.P. and CryoGARS (2014). Water in snow likes to go with the flow: Dynamics of liquid water in snow and its impact on stability. <i>Proceedings of the 2014 International Snow Science Workshop</i>
	Dean, J., H.-P. Marshall , Lutz, E., Christian, J., Markle, B., and Whitmore, S. (2014). Using nir reflectance for characterizing stratigraphy and liquid water in snow. <i>Proceedings of the 2014 International Snow Science Workshop</i>

- 2014 Elder, K., **H.-P. Marshall**, Elder, L., Starr, B., Karlson, A., and Robertson, J. (2014). Design and installation of a tipping bucket snow lysimeter. *Proceedings of the 2014 International Snow Science Workshop*
- Havens, S., **H.-P. Marshall**, Johnson, J., and Nicholson, B. (2014a). Calculating the velocity of a fast moving snow avalanche using an infrasound array. *Proceedings of the 2014 International Snow Science Workshop*
- Hedrick, A. and **H.-P. Marshall** (2014). Automated snow depth measurements in avalanche terrain using time-lapse photography. *Proceedings of the 2014 International Snow Science Workshop*
- Lutz, E. and **H.-P. Marshall** (2014). Validation study of avatech's rapid snow penetrometer, sp1. *Proceedings of the 2014 International Snow Science Workshop*
- Robertson, M., **H.-P. Marshall**, Roberts, D., and Tidd, W. (2014). Measuring snow water equivalent in wet snow with ultrawideband gpr. *Proceedings of the 2014 International Snow Science Workshop*
- Rodriguez, S., **H.-P. Marshall**, and Rodriguez, P. (2014). Applications of low cost and low power fmcw radar in the snow characterization of dry snow. *Proceedings of the 2014 International Snow Science Workshop*
- 2012 **H.-P. Marshall**, , Gleason, A., Heilig, A., Koh, G., and Sturm, M. (2012). Spatial variability of snow stratigraphy from microwave radar. *Proceedings of the 2012 International Snow Science Workshop*
- Havens, S., **H.-P. Marshall**, Covington, C., Nicholson, B., and Conway, H. (2012). Real time slope stability modeling of direct avalanches. *Proceedings of the 2012 International Snow Science Workshop*
- Hedrick, A., **H.-P. Marshall**, Heilig, A., and Winstral, A. (2012). Testing and implementing a wind redistribution model in mountainous terrain using a high-resolution LiDAR-derived DEM. *Proceedings of the 2012 International Snow Science Workshop*
- Heilig, A., **H.-P. Marshall**, and Evans, S. (2012). Temporal and spatial changes of a seasonal snowpack on a small sheltered slope. *Proceedings of the 2012 International Snow Science Workshop*
- 2008 **H.-P. Marshall**, Birkeland, K., Elder, K., and Meiners, T. (2008a). Helicopter-based microwave radar measurements in alpine terrain. *Proceedings of the 2008 International Snow Science Workshop*
- H.-P. Marshall**, Landry, C., Hale, S., Roberts, J., and Conway, H. (2008c). Snow slope stability modeling of direct-action avalanches in a continental climate: Red mountain pass, co. *Proceedings of the 2008 International Snow Science Workshop*

- 2008 **H.-P. Marshall**, Koh, G., and Sturm, M. (2008b). Ultra-broadband portable microwave fmcw radars for measuring snow depth, snow water equivalent, and stratigraphy: practical considerations. *International Union of Radio Science (URSI) General Assembly*. (Invited)
- Pielmeier, C. and **H.-P. Marshall** (2008). Estimating rutschblock stability from snowmicropen measurements. *Proceedings of the 2008 International Snow Science Workshop*
- Gleason, A., **H.-P. Marshall**, and McCreight, J. (2008). Into the belly of the beast (snow depth across starting zones measured with frequency modulated continuous wave radar). *Proceedings of the 2008 International Snow Science Workshop*
- Birkeland, E. L. K., **H.-P. Marshall**, and Hansen, K. (2008). Quantifying changes in weak layer microstructure associated with loading events. *Proceedings of the 2008 International Snow Science Workshop*
- 2007 **H.-P. Marshall**, Rutter, N., Koh, G., O'Neel, S., and McCreight, J. (2007). Ground-based fmcw radar measurements of dry snowpacks during the 2006-07 nasa clpx field experiment. *Eastern Snow Conference*
- Demuth, M., **H.-P. Marshall**, Morris, E., Burgess, D., and Bell, C. (2007). High-resolution near-surface snow stratigraphy inferred from ground-based broadband microwave radar measurements: Devon ice cap, nunavut, canada 2005-06, cryosat/asiras validation campaign. *Eastern Snow Conference*
- 2006 Demuth, M., Morris, E., **H.-P. Marshall**, Fisher, D., Sekerka, J., Koerner, R., and Gray, L. (2006). Characterizing percolation and wet-snow facies variability on devon ice cap, nunavut, canada. *Proceedings of the 2006 International Snow Science Workshop*
- H.-P. Marshall**, Koh, G., Sturm, M., Johnson, J., Demuth, M., Landry, C., Deems, J., and Gleason, A. (2006a). Spatial variability of the snowpack: experiences with measurements at a wide range of length scales with several different high precision instruments. *Proceedings of the 2006 International Snow Science Workshop*
- Gleason, A. and **H.-P. Marshall** (2006). Deformation analysis of the snow micropen using particle image velocimetry. *Proceedings of the 2006 International Snow Science Workshop*
- Pielmeier, C., Schweizer, J., and **H.-P. Marshall** (2006). Improvements in the application of the snowmicropen to derive stability information for avalanche forecasting. *Proceedings of the 2006 International Snow Science Workshop*
- 2005 Koh, G. and **H.-P. Marshall** (2005). Radar signature of recently disturbed soil for mine detection. *SPIE - International Society for Optical Engineering, Session on Detection and Remediation Technologies for Mines and Mine like Targets*
- 2004 Hayes, P., Wilbour, C., Gibson, R., **H.-P. Marshall**, and Conway, H. (2004). A simple model of snow slope stability during storms. *Proceedings of the 2004 International Snow Science Workshop*

- 2004 **H.-P. Marshall**, Schneebeli, M., Koh, G., Matzl, M., and Pielmeier, C. (2004). Measurements of snow stratigraphy with fmcw radar: comparison with other snow science instruments. *Proceedings of the 2004 International Snow Science Workshop*
- 2002 **H.-P. Marshall** (2002a). Modeling the interface between two layers of snow. *Proceedings of the 2002 International Snow Science Workshop*
- 1998 **H.-P. Marshall**, Conway, H., and Rasmussen, L. (1998). Snow densification during rain. *Proceedings of the 1998 International Snow Science Workshop*

AGU Talks

- 2015 McNamara, J. P., **Marshall, H.P.**, Kohn, M., Evans, S., and Flores, A. (2015). Lateral flow in snow as a runoff generation mechanism. *Fall Meeting, AGU, San Francisco, CA, Abstract C44B-01*
- Lewis, G., Hawley, R., Osterberg, E., and **Marshall, H.P.** (2015). Icebridge radar as a tool for understanding accumulation variability throughout the western greenland ice sheet. *Fall Meeting, AGU, San Francisco, CA, Abstract C51E-04*
- Deeb, E., **Marshall, H.P.**, Forster, R. R., Arcone, S., Siqueira, P., and Morriss, B. (2015). The present and future of using insar to estimate swe. *Fall Meeting, AGU, San Francisco, CA, Abstract C31B-04*
- 2014 **Marshall, H.P.**, Heilig, A., Evans, S., Robertson, M., Hetrick, H., Eiriksson, D., Dean, J., Karlson, A., Hedrick, A., Bradford, J. H., McNamara, J. P., Flores, A., Kohn, M., and Rodriguez, C. (2014). Liquid water dynamics in unsaturated snow: the role of lateral flow. *Fall Meeting, AGU, San Francisco, CA, Abstract C22B-07*
- Hedrick, A., **Marshall, H.P.**, Winstral, A., Elder, K., Yueh, S., and Cline, D. (2014b). Using regional-scale lidar surveys to validate operational snow models. *Fall Meeting, AGU, San Francisco, CA, Abstract C32A-06*
- Deeb, E., **Marshall, H.P.**, Lamie, N., and Arcone, S. (2014). Quantifying the accuracy of snow water equivalent estimates using broadband radar signal phase. *Fall Meeting, AGU, San Francisco, CA, Abstract C31E-05*
- Havens, S., **Marshall, H.P.**, Trisca, G., Johnson, J. B., and Nicholson, B. (2014d). Modeling and monitoring avalanches caused by rain-on-snow events. *Fall Meeting, AGU, San Francisco, CA, Abstract C31E-03*
- McNamara, J. P., Aishlin, P., Flores, A., Benner, S. G., **Marshall, H.P.**, and Pierce, J. (2014). Integrating local research watersheds into hydrologic education: Lessons from the dry creek experimental watershed. *Fall Meeting, AGU, San Francisco, CA, Abstract ED54A-3508*

- 2013 Robertson, M., Koenig, L., Trisca, G., and **Marshall, H.P.** (2013). Accumulation mapping at summit, greenland using an autonomous rover (invited). *Fall Meeting, AGU, San Francisco, CA, Abstract C24B-07*
- 2012 Gusmeroli, A., Wolken, G., Arendt, A. A., Campbell, S., O'Neel, S., and **Marshall, H.P.** (2012). Ground-penetrating radar observations of winter snow accumulation on alaska glaciers. (invited). *Fall AGU, San Francisco, CA, Abstract C52A-03*
- 2011 Alexander, P., Tedesco, M., Steiner, N., **H.-P. Marshall**, Luthcke, S. B., and Fettweis, X. (2011). Identification of accumulation, density and grain size bias in the regional climate model mar over the greenland ice sheet using in-situ and remotely sensed data. *Fall Meeting, AGU, San Francisco, CA, Abstract C22B-06*
- 2010 **H.-P. Marshall**, Marks, D. G., and Hudak, A. H. W. R. S. A. T. (2010). Estimating swe distribution with a combination of ground-based radar measurements, modeling and remote sensing. *Fall AGU, San Francisco, CA, Abstract C11C-01*. (Invited)
- Yueh, S. H., Rott, H., Nagler, T. F., Cline, D. W., Duguay, C. R., Essery, R., Etchevers, P., Hajnsek, I., Kern, M., Macelloni, G., Malnes, E., Pulliainen, J. T., Tsang, L., Xu, X., **H.-P. Marshall**, and Elder, K. (2010). Microwave radar retrieval of snow water equivalent. *Fall Meeting, AGU, San Francisco, CA, Abstract C54B-03*
- Bradford, J. H. and **H.-P. Marshall** (2010). Estimating complex dielectric permittivity of soils from spectral ratio analysis of swept frequency (FMCW) ground-penetrating radar data. *Fall Meeting, AGU, San Francisco, CA, Abstract H21K-06*. (Invited)
- Johnson, J. B., Marcillo, O. E., Arechiga, R. O., Johnson, R., Edens, H. E., **H.-P. Marshall**, Havens, S., and Waite, G. P. (2010). Investigation of the infrasound produced by geophysical events such as volcanoes, thunder, and avalanches: the case for local infrasound monitoring. *Fall Meeting, AGU, San Francisco, CA, Abstract S14A-01*. (Invited)
- 2009 **H.-P. Marshall**, Demuth, M. N., Gray, L., Burgess, D. O., and Morris, E. M. (2009a). High-resolution annual accumulation rate measurements using FMCW radar, Devon Ice Cap, Nunavut, Canada. *EOS Trans. AGU Fall meeting suppl., Abstract C34B-01*, 90. (Invited)
- McCreight, J. L., Rajagopalan, B., Slater, A. G., and **H.-P. Marshall** (2009). A comparison of methods for estimation and prediction of snow depth and its spatial non-stationarity at the first-order basin scale. *EOS Trans. AGU Fall meeting suppl., Abstract C21E-06*, 90. (Invited)
- Lutz, E., Birkeland, K., **H.-P. Marshall**, and Hansen, K. (2009a). Spatial and temporal analysis of snowpack strength and stability and associations with micro-climate on an inclined forest opening. *EOS Trans. AGU Fall meeting suppl., Abstract C21E-08*, 90. (Invited)
- 2008 **H.-P. Marshall**, Rutter, N., Tape, K., Sturm, M., and Essery, R. (2008d). High resolution ground-based snow measurements during the NASA CLPX-II campaign, north slope, alaska. *EOS Trans. AGU Fall meeting suppl., Abstract C12B-05*, 89(53)

- 2008 Rutter, N., **H.-P. Marshall**, Tape, K., and Essery, R. (2008). Observing microscale variations in snow stratigraphy using near infra-red photography. *EOS Trans. AGU Fall meeting suppl., Abstract C34A-07*, 89(53). (Invited)
- 2007 **H.-P. Marshall**, Koh, G., Sturm, M., and Rutter, N. (2007c). Estimating snow depth, snow water equivalent, and stratigraphy at high resolution using microwave radar. *EOS Trans. AGU Fall meeting suppl., Abstract H23H-04*, 88(52). (Invited)
- Matsuoka, K. and **H.-P. Marshall** (2007). Laboratory and numerical experiments of radar back scattering: towards interpreting the shape of subsurface targets in cryospheric applications. *EOS Trans. AGU Fall meeting suppl., Abstract NS14A-04*, 88(52). (Invited)
- 2006 **H.-P. Marshall**, Sturm, M., Holmgren, J., and Koh, G. (2006b). Ground-based fmcw radar measurements for calibration/validation during amsr-ice06: Barrow, ak, usa. *EOS Trans. AGU Fall meeting suppl., Abstract C13A-07*, 87(52)
- 2005 McNamara, D., O’Neel, S., **H.-P. Marshall**, and Pfeffer, T. (2005). Iceberg calving at rapidly retreating tidewater glacier: Columbia glacier, ak, usa. *EOS Trans. AGU Fall meeting suppl., Abstract C51A-0261*, 86(52)
- 2003 **H.-P. Marshall**, Koh, G., and Forster, R. (2003). Ground-based fmcw radar measurements: a summary of the nasa clpx data. *EOS Trans. AGU Fall meeting suppl., Abstract C42B-07*, 84(46)
- 2001 Pfeffer, T., **H.-P. Marshall**, Harper, J., and Humphrey, N. (2001). The constitutive relationship of a glacial-scale block of temperate ice. *EOS Trans. AGU Fall meeting suppl., Abstract IP12B-04*, 82(47)

AGU Posters

- 2014 Marks, D., **Marshall, H.P.**, Winstral, A., and Hedrick, A. (2014). The sensitivity of mountain snowcovers to temperature, humidity, and phase change in a warming climate. *Fall Meeting, AGU, San Francisco, CA, Abstract C43E-0438*
- Hedrick, A., Marks, D., Winstral, A., and **Marshall, H.P.** (2014a). Evaluating mesoscale numerical weather predictions and spatially distributed meteorologic forcing data for developing accurate swe forecasts over large mountain basins. *Fall Meeting, AGU, San Francisco, CA, Abstract C43D-0431*
- Robertson, M. and **Marshall, H.P.** (2014). Ground penetrating radar for measuring snow water equivalent in wet snow. *Fall Meeting, AGU, San Francisco, CA, Abstract C43D-0426*
- 2013 Havens, S., Johnson, J. B., **Marshall, H.P.**, Nicholson, B., and Trisca, G. (2013a). Snow avalanche detection and identification for near real-time application. *Fall AGU, San Francisco, CA, Abstract S23B-2493*

- 2013 Hetrick, H., **Marshall, H.P.**, Bradford, J. H., and Mead, J. (2013). Ert inversion with the incorporation of surface-based gpr. *Fall Meeting, AGU, San Francisco, CA, Abstract NS33A-1686*
- Watson, K. A., **Marshall, H.P.**, McNamara, J., and Flores, A. (2013). Influence of high-resolution land-surface initialization on near surface hydrometeorological variables in a coupled land-atmosphere model. *Fall Meeting, AGU, San Francisco, CA, Abstract H21A-1011*
- Tinkham, W. T., Smith, A., **Marshall, H.P.**, Falkowski, M. J., and Winstral, A. (2013b). Quantifying spatial distribution of snow depth errors from lidar using random forests. *Fall Meeting, AGU, San Francisco, CA, Abstract H13J-1497*
- Marshall, H.P.**, Sturm, M., Elder, K., and Yueh, S. (2013). Extrapolating from pits: Quantifying spatial variability of snow for remote sensing calibration/validation (invited). *Fall Meeting, AGU, San Francisco, CA, Abstract C41C-0650*
- Dean, J., **Marshall, H.P.**, Rutter, N., and Karlson, A. (2013). Improving nir snow pit stratigraphy observations by introducing a controlled nir light source. *Fall Meeting, AGU, San Francisco, CA, Abstract C41C-0646*
- Aguayo, M., **Marshall, H.P.**, McNamara, J. P., Mead, J., and Flores, A. (2013). Reconstruction of layer densities in a multilayer snowpack using a bayesian approach to inverse modeling. *Fall Meeting, AGU, San Francisco, CA, Abstract C41A-0579*
- Hedrick, A., **Marshall, H.P.**, Elder, K., and Yueh, S. (2013). Using ground truth-validated, large-scale airborne lidar to assess the accuracy of the snodas hydrological model in complex terrain. *Fall Meeting, AGU, San Francisco, CA, Abstract C41B-0622*
- van Wijk, K., Otheim, L., **Marshall, H.P.**, Kurbatov, A., and Spaulding, N. (2013). laser ultrasonic characterization of ice cores. *Fall Meeting, AGU, San Francisco, CA, Abstract C13C-0694*
- Trisca, G., Robertson, M., **Marshall, H.P.**, Koenig, L., and Comberiate, M. (2013). Grover: An autonomous vehicle for ice sheet research. *Fall Meeting, AGU, San Francisco, CA, Abstract C13C-0691*
- 2012 Kormos, P. R., McNamara, J. P., Seyfried, M. S., Marks, D., Flores, A., **Marshall, H.P.**, and Williams, C. J. (2012). Critical zone soil properties effects on soil water storage and flux. *Fall AGU, San Francisco, CA, Abstract H53I-1656*
- Deeb, E., **Marshall, H.P.**, LeWinter, A., Finnegan, D., Deems, J. S., and Landry, C. (2012). Comparison of coordinated satellite and ground-based x-band radar collections for the retrieval of snow parameters. *Fall AGU, San Francisco, CA, Abstract C33C-0681*
- Evans, S., Heilig, A., Kohn, M., and **Marshall, H.P.** (2012). Isotopic dynamics in a seasonal snowpack. *Fall AGU, San Francisco, CA, Abstract C33C-0673*

- 2011 Hopkins, A. J., Heilig, A., **H.-P. Marshall**, and Flores, A. N. (2011). Potential application of nasa smap radar data to detect the aerial extent of snowpack melt. *Fall AGU, San Francisco, CA, Abstract H23G-1360*
- Havens, S. and **H.-P. Marshall** (2011). Comparison of snowmicropenetrometer estimated microstructure and frequency modulated continuous wave radar. *Fall AGU, San Francisco, CA, Abstract C31A-0600*
- Lutz, E., Hawley, R. L., **H.-P. Marshall**, Osterberg, E. C., Courville, Z., Overly, T. B., and Wong, G. J. (2011). Thule to summit: Understanding the physical properties of northern greenland near-surface snow and firn. *Fall AGU, San Francisco, CA, Abstract C33A-0624*
- Walters, R. D., McNamara, J. P., **H.-P. Marshall**, and Flores, A. N. (2011). A physically-based approach to downscale coarse snow model output for hydrologic modeling at hillslope scales. *Fall AGU, San Francisco, CA, Abstract C33D-0668*
- Tinkham, W. T., Hoffman, C. M., Falkowski, M. J., Smith, A. M., Link, T. E., and **H.-P. Marshall** (2011). Spatial accounting for errors in lidar-derived products: Snow volume and snow water equivalent estimation. *Fall AGU, San Francisco, CA, Abstract C33D-0672*
- Deeb, E. J., **H.-P. Marshall**, Finnegan, D. C., Deems, J. S., and Landry, C. (2011). Comparison of ground-based lidar and ground-based radar of southwestern colorado snowpack. *Fall AGU, San Francisco, CA, Abstract C33D-0673*
- H.-P. Marshall**, Deeb, E. J., Gleason, A., Heilig, A., Finnegan, D. C., Deems, J. S., Havens, S., Kormos, P. R., Landry, C., and McCreight, J. L. (2011). Alpine snow distribution from-ground based radar measurements compared with a high resolution digital elevation model from ground-based lidar observations. *Fall AGU, San Francisco, CA, Abstract C33D-0674*
- Heilig, A., **H.-P. Marshall**, Mayer, C., and Winstral, A. H. (2011). Spatial variability of SWE in alpine areas - how do variability patterns change with grid designs? *Fall AGU, San Francisco, CA, Abstract C33E-0686*. (Invited)
- Eiriksson, D., McNamara, J. P., **H.-P. Marshall**, and Bradford, J. H. (2011). Assessment of the hydrologic significance of lateral water flow in snow. *Fall AGU, San Francisco, CA, Abstract C33E-0688*
- Kormos, P. R., McNamara, J. P., Marks, D. G., Flores, A. N., **H.-P. Marshall**, and Boe, E. (2011). Use of plot scale observations to gauge the applicability of physically-based models. *Fall AGU, San Francisco, CA, Abstract H33H-1418*
- Bradford, J. H., **H.-P. Marshall**, and Dickins, D. (2011). Modeling frequency dependent gpr wave propagation through sea ice using the reflectivity method and application to oil-spill detection. *Fall AGU, San Francisco, CA, Abstract NS51A-1742*

- 2010 Rutter, N., **H.-P. Marshall**, Tape, K. D., and Essery, R. (2010). Observations of snow-pack properties to evaluate ground-based microwave remote sensing. *Fall Meeting, AGU, San Francisco, CA, Abstract C33A-0513*
- Havens, S., **H.-P. Marshall**, Elder, K., Rutter, N., and Tape, K. D. (2011). Snow micro penetrometer for classifying grain types in the alpine and arctic environment. *Fall Meeting, AGU, San Francisco, CA, Abstract C33A-0514*
- Tedesco, M., **H.-P. Marshall**, and Steiner, N. (2010). From colorado to greenland: the 2010 ground passive and active snow (gaps) experiment. *Fall AGU, San Francisco, CA, Abstract C33A-0516*
- Shallcross, A. T. J. M. A. F. H. M. D. M. N. G. (2010). Estimating basin snow volume using aerial lidar and binary regression trees. *Fall Meeting, AGU, San Francisco, CA, Abstract C33C-0537*. (Invited)
- Deems, J. S., Finnegan, D. C., Deeb, E. J., **H.-P. Marshall**, Bryant, A. C., Skiles, S., Landry, C., and Painter, T. H. (2010). A comparison of ground-based LiDAR, contact spectroscopy, FMCW radar, and manual snow pit profiles of a mountain snowpack. *Fall Meeting, AGU, San Francisco, CA, Abstract C33D-0562*
- Deeb, E. J., **H.-P. Marshall**, Deems, J., Finnegan, D. C., Painter, T. H., Landry, C., Bryant, A. C., and Skiles, S. (2010). Field collection efforts of snowpack properties in support of remote sensing applications. *Fall Meeting, AGU, San Francisco, CA, Abstract C33D-0564*
- Boe, E., McNamara, J. P., and **H.-P. Marshall** (2010). Quantifying snow variability using an inexpensive network of ultrasonic depth sensors. *Fall Meeting, AGU, San Francisco, CA, Abstract H11A-0796*
- 2009 Steiner, N., Tedesco, M., **H.-P. Marshall**, and Josberger, E. G. (2009). The 2009 Ground Passive and Active Snow (GAPS) Experiment. *EOS Trans. AGU Fall meeting suppl., Abstract C51B-0483*, 90
- Anderson, B., McNamara, J. P., and **H.-P. Marshall** (2009). Sources of snow cover variability in complex terrain. *EOS Trans. AGU Fall meeting suppl., Abstract C31D-0461*, 90
- H.-P. Marshall**, Koh, G., and Sturm, M. (2009b). A tool for rapid snow depth, stratigraphy, and swe measurements: portable microwave radar. *EOS Trans. AGU Fall meeting suppl., Abstract C31D-0465*, 90
- Shallcross, A. T., McNamara, J. P., **H.-P. Marshall**, Marks, D. G., Link, T. E., and Winstral, A. H. (2009). Estimating snow volume in mountain catchments using aerial LiDAR. *EOS Trans. AGU Fall meeting suppl., Abstract C31D-0468*, 90

- 2009 Demuth, M. N., **H.-P. Marshall**, Morris, E. M., Burgess, D. O., and Gray, L. (2009). The variation of polar firn subject to percolation - characterizing processes and glacier mass budget uncertainty using high-resolution instruments. *EOS Trans. AGU Fall meeting suppl.*, C31E-0470, 90
- 2008 Lutz, E., Birkeland, K., and **H.-P. Marshall** (2008). Important factors influencing microstructural estimates derived from the snowmicropen. *EOS Trans. AGU Fall meeting suppl.*, Abstract C21C-0579, 89(53)
- McCreight, J., Slater, A., **H.-P. Marshall**, and Deems, J. (2008). Non-stationarity and statistical partitioning of snow depth fields: Independent variable controls and application to sampling design and modeling. *EOS Trans. AGU Fall meeting suppl.*, Abstract H23B-0963, 89(53)
- 2007 **H.-P. Marshall**, Gleason, A., Landry, C., and McCreight, J. (2007d). Snow depth and snow water equivalent distribution in a high alpine basin: quantifying the length scales and magnitude of variation in senator beck basin, colorado. *EOS Trans. AGU Fall meeting suppl.*, Abstract GC41A-0104, 88(52)
- Deeb, E., Forster, R., **H.-P. Marshall**, and Rutter, N. (2007). Ground-based radar measurements of the northern colorado snowpack at clpx- ii. *EOS Trans. AGU Fall meeting suppl.*, Abstract C23A-0929, 88(52)
- 2006 Schneebeli, M., **H.-P. Marshall**, and Johnson, J. (2006). Snow properties considered from a multiscale perspective: new measurement techniques. *EOS Trans. AGU Fall meeting suppl.*, Abstract C21C-1163, 87(52). (Invited)
- 2003 Gleason, J. and **H.-P. Marshall** (2003). Measurements of snow deformation using particle image velocimetry combined with finite element simulations. *EOS Trans. AGU Fall meeting suppl.*, Abstract C41B-0952, 84(46)
- 2002 **H.-P. Marshall** (2002b). Modeling the interface between two layers of snow using finite elements. *EOS Trans. AGU Fall meeting suppl.*, Abstract C11B-0996, 83(47)
- Waddington, E., Neumann, T., Morse, D., and **H.-P. Marshall** (2002). Spatial accumulation-rate pattern inferred from radar internal layers and point measurements of velocity and accumulation near taylor mouth, victoria land. *EOS Trans. AGU Fall meeting suppl.*, Abstract C12A-0998, 83(47)
- 2001 Harper, J., Bradford, J., **H.-P. Marshall**, Gleason, J., Pfeffer, T., and Humphrey, N. (2001). Snow stratigraphy in an alpine environment: Spatial variability and a comparison of measurement methods. *EOS Trans. AGU Fall meeting suppl.*, Abstract IP51A-0740, 82(47)

BSU Undergraduate Research Symposium, Student Presentations

- 2012 | Burch, K., Drake, K., II, L. L., **H.-P. Marshall**, and Planting, A. (2012). Snow penetrometer embedded system design. *BSU Undergraduate Research and Scholarship Conference*
- Peterson, A. and **H.-P. Marshall** (2012). Estimates of snow accumulation rates over the greenland ice sheet using airborne radar. *BSU Undergraduate Research and Scholarship Conference*
- Robertson, J. and **H.-P. Marshall** (2012). Snow melt dynamics. *BSU Undergraduate Research and Scholarship Conference*

Computer Programming Skills

Windows, DOS, Macintosh, Unix, Linux operating systems

Matlab (Data Acq., Signal Proc., Statistics, PDE, Wavelet, Symbolic Math Toolboxes),

Developed GUI-driven analysis software for FMCW radar acquisition and processing, snowpit data display, and seismic signal detection of ice quakes.

C++, LaTeX, Python, Abaqus, IDL, GPSurvey, LabView, Awk, GMT, Pine, Emacs, Vi

Campbell Scientific data logger communications programs

S+, Mathematica, Surfer, IslandDraw, Excel, Access, Visual Basic

data acquisition, data analysis, signal processing, database management, finite element analysis

low power microcontrollers for ADC/DAC and data logging

Skills / Certifications

Ground Penetrating Radar (monopulse and FMCW, 5 MHz - 18 GHz)

GPS + theodolite survey equipment, SnowMicroPenetrometer

Northwest Avalanche Institute - Level 1 certification (beginning)

Northwest Avalanche Institute - Level 2 certification (advanced)

American Institute for Avalanche Research and Education - Level 1 Instructor Training

CPR for the Professional rescuer

First Responder (EMT program, North Seattle C.C.)

Snow survival and crevasse rescue - U.S. Antarctic Program

experience with snowmobile operation and repair

experience with electronic, metal, and wood-working tools

geophysical instrument design and fabrication

Volunteer Activities

Special Olympics basketball coach (02/15-06/15)
ESL Teacher for refugees (08/09-07/10)
Mentor, Big Brothers / Big Sisters of La Plata County (07/2006-08/2008)
Volunteer high school teacher, Old Crow, Yukon Territory, Canada (1999-2000)
Helped install seismic station on the summit of Mt. Rainier (1999)
Logan School of Learning (2-3rd graders), Field Trip leader to Loveland Pass, CO (2003)
Co-taught 3-day English course to indigenous leaders in southern Chile (2003)
Guest lecturer, Nathan Hale High School Physics class (1999)
Notetaker for Disabled Student Services, University of Washington (1996)
Teen Feed (1995)

Affiliations

Member, Snow Radar Algorithm Team, NASA SCLP (2004-present)
International Glaciological Society (IGS)
American Geophysical Union (AGU, Cryosphere Representative 2008-2012)
Sigma Xi (Full member)
American Avalanche Association (Research Chair 2005-2011, Professional Member)
American Institute for Avalanche Research and Education (AIARE) Certified Instructor (2003-2004)
American Meteorological Society (Full Member)

Interests / Hobbies

Backcountry skiing / snowboarding, mountaineering, rock climbing, basketball, trail running
West African percussion, Flamenco percussion, yoga